

Assignment

PART A

1) $255.375_{10} \rightarrow$ Binary, Octal, Hexadecimal

a)

	255	Remainders
2	255	
2	127	1
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	0

11111111.011_2
 255.375

$0.375 \times 2 = 0.75 \rightarrow 0$
 $0.75 \times 2 = 1.5 \rightarrow 1$
 $0.5 \times 2 = 1 \rightarrow 1$

OR $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$
 $1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1$

b) 01111111.011_2
 377.3_8

c) 11111111.011_2

10 11 12 13 14 15
A B C D E F

FF.6₁₆

2) Reverse Thinking :- $\frac{1 \times 1}{2^1} + \frac{0 \times 1}{2^2} + \frac{1 \times 1}{2^3} = 0.5 + 0 + 0.125$
 $= 0.625$

110101
 $2^5 + 2^3 + 2^2 + 2^0 \Rightarrow 32 + 16 + 4 + 1 = 53$
 53.625

3) 100111
 654321

$0+1+1=2 \rightarrow$ odd positions $1+0+1=2$ even positions
 Difference between even and odd positions should be divisible by 3.

4) Basic procedure: 1) Take one's complement } Negative Numbers
 2) Then add one

Example: -23

00010111
 1's complement $\rightarrow 11101000$, Now add 1, 11101000
 $+ 1$
 11101001 $\Rightarrow -23_{10}$

5) $(A+B)(A'+C)(B+C')$ properties reduced

$[A'A' = 0]$

$= [A'A' + A'C + A'B + BC] (B+C')$

$[B'B = B]$

$= ABC + A'B'C' + A'B'C + A'B'C'$

$= ABC + BC + A'B + A'B'C'$

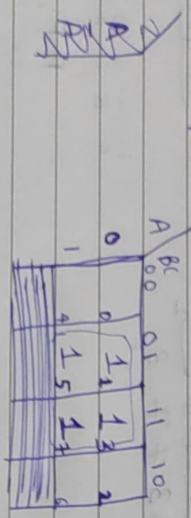
$= BC(A+1) + A'B(1+C')$

$= B(C+A')$

$\begin{matrix} X+1 = 1 \\ A+1 = 1 \\ 1+C' = 1 \end{matrix}$

I have simplified the terms and eliminated as well using properties (written in boxes)

6)



→ C

Reason we use properties, as it reduces to C.

↓, AB A & B changes, we will not consider it, because

Code: module xor_gate (input a, input b, output y);

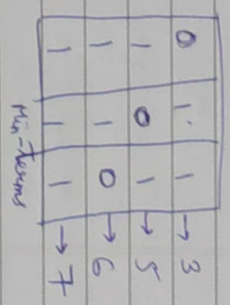
assign y = a^b;

endmodule

PART C: (will try to make Truth Table)

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

K-Map



Output = $BC + AC + AB$

2) Code

module andgate (

input a, // first input

input b, // second input

output y);

// perform AND operation

assign y = a & b;

endmodule

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1